TEXORA · LECTURE SHEET

3D Models of Fiber, Yarn, Fabric and Apparel

Modelling the textile world itself: helical fibre assemblies and the geometry of twist, yarn and fabric unit cells in CAD and Blender (curves, screw and array tools, cloth simulation), and virtual apparel prototyping.

ME-1202 · ENGINEERING DRAWING-II · TOPIC 3 · LEVEL 1 TERM 2

1. Why model textiles?

Textile structures are geometry all the way down — helices in yarns, interlaced waves in fabrics — so they repay exact 3D modelling: predicting air permeability and filtration from pore geometry, estimating fabric mechanics before a loom is threaded, visualising finishes, and prototyping garments without sewing a single sample. The modeller's toolkit in SOLIDWORKS and Blender maps neatly onto textile hierarchy: a fibre is a swept tube, a yarn is a bundle of helices, a fabric is an array of interlaced unit cells, a garment is a cloth simulation over a body.

2. Fibre and yarn models — the geometry of twist

A single fibre is modelled as a sweep: its true cross-section (round for most synthetics, kidney-shaped and convoluted for cotton, trilobal for lustre nylon) carried along a path. A **spun yarn** is built by sweeping fibre cross-sections along **helices** about the yarn axis — in Blender, a curve given thickness (bevel) and twisted with the **screw modifier**; in CAD, a helix-based sweep. Twist is geometry with a formula: for yarn diameter D and twist T (turns per unit length), the surface fibre makes the **twist angle**

$$\tan \theta = \pi D T$$

WORKED EXAMPLE

Yarn D = 0.20 mm, twist 800 turns/m (0.80 turns/mm): $\tan \theta = \pi \times 0.20 \times 0.80 = 0.503 \rightarrow \theta \approx$ **26.7°**.

One turn of that surface helix: pitch = 1/0.80 = 1.25 mm of yarn length, circumference = πD = 0.628 mm, so fibre length per turn = $\sqrt{(1.25^2 + 0.628^2)}$ = **1.40 mm** — the fibre travels 1.40 mm to advance the yarn 1.25 mm (about 12% "extra path", the geometric root of yarn contraction and of why twisted yarn is springier than its fibres).

A ring of such helices at several radii, with fibre migration ignored for a first model, already looks and behaves like yarn in section; adding a core-and-wrap arrangement models core-spun yarns, and two opposed helix families model plied yarn.

3. Fabric models — the woven unit cell

A woven fabric repeats a **unit cell**: one warp yarn and one weft yarn interlacing by the weave plan. For plain weave the yarns undulate over-under-over; their centre-lines are modelled as sine-like curves (Peirce's cloth geometry, the classical analytical version, treats them as circular arcs and straight segments), swept with flattened (lenticular) cross-sections because yarns squash where they press. In practice: model one unit cell with curves + bevel in Blender (or sweep in CAD), then tile it with the **array modifier** in warp and weft directions; twill and satin cells just lengthen the float pattern. Knitted fabrics replace interlacement with intermeshed loops — a loop is one repeated curve, arrayed in courses and wales. Research codes (TexGen, WeavePoint-style tools) generate these cells from weave data automatically and export them to FEA, where pore size predicts permeability (Topic 2's simulation, textile edition). The **solidify modifier** gives the fabric sheet its thickness for rendering and printing.

4. Apparel models — cloth on a body

Garments are modelled two ways. **Pattern-based virtual prototyping** (CLO 3D, Browzwear, and Blender cloth workflows) sews flat 2D pattern pieces together *in simulation* around a digital avatar: the cloth solver stretches, bends and collides until the garment drapes as physics dictates. Inputs that matter: fabric bending and shear stiffness (from KES or FAST measurements), friction, and the avatar's measurements. The output — fit maps showing tension and ease, true-to-life renders for buyers — lets a factory in Gazipur approve a style with a brand in Europe without air-freighting samples, which is why virtual sampling has become standard practice in the export industry. **Direct 3D design** (sculpting on an avatar) suits concept work and knitwear.

5. Key points

- Hierarchy: fibre = swept true cross-section; yarn = helices about an axis; fabric = arrayed unit cell; garment = simulated cloth.
- Twist angle $\tan \theta = \pi DT$; the surface fibre's helical path is longer than the yarn it builds (contraction).
- Plain-weave cell: interlaced undulating yarns, flattened in section; arrays tile the cell into cloth; knits array a loop instead.
- Cloth simulation needs measured bending/shear properties; its industrial prize is sample-free fit approval.

6. Practice

Answer in your own words; brief model answers follow.

1. A yarn of diameter 0.25 mm is twisted at 600 turns/m. Find the surface twist angle.
2. Describe, step by step in Blender terms, how you would build a short length of two-ply yarn.
3. Why is a cotton fibre's model not just a smooth round tube, and what two visual properties does its real cross-section explain?
4. Sketch (in words) the unit cell of a 2/1 twill and state how it differs from the plain-weave cell when both are arrayed.
5. A virtual fit session shows red tension bands across the chest of a shirt on the avatar. Name two pattern changes that relieve them, and the cloth property whose wrong value could make the simulation lie about it.
6. Why do fabric models destined for permeability simulation keep the true pore openings, while models for a sales render may cheat them closed?

Answer key

1. T = 600 turns/m = 0.60 turns/mm. $\tan\theta = \pi \times 0.25 \times 0.60 = 0.471 \rightarrow \theta \approx 25.2°$. (Finer yarn or lower twist reduces θ; compare 26.7° for the worked example.)
2. (1) Draw one fibre path as a helix — a curve wrapped about the yarn axis, or a straight curve given a screw modifier with the yarn's twist pitch. (2) Give it thickness with bevel/curve fill and a fibre cross-section profile. (3) Repeat fibres at several radii (duplicate + rotate / array about the axis) to build one single yarn. (4) Duplicate the whole yarn, mirror its helix handedness as required, and offset the two singles side by side with a coarser opposite-direction screw — plying twist runs opposite to singles twist (S on Z) for a balanced two-ply. (5) Apply soft-body-safe materials: slight colour variation per fibre sells the realism.
3. Cotton's cross-section is a collapsed, kidney-shaped ribbon with a central lumen, and the fibre twists in convolutions along its length. The model affects lustre (a ribbon scatters light diffusely — cotton's soft matt look, unlike a round filament's sparkle) and absorbency/spinning behaviour (the convoluted ribbon grips neighbouring fibres and holds water in its lumen and surface folds). A round-tube model renders like a synthetic filament and misleads both the eye and any surface-area calculation.
4. Plain weave cell: one warp yarn passes over one weft, under the next — maximum interlacement, symmetrical undulation. A 2/1 twill cell: the warp floats over two wefts then under one, and the pattern steps sideways each row, so the arrayed model shows diagonal twill lines, longer floats, a smoother face and less crimp. Its unit cell spans 3 × 3 yarn positions instead of 2 × 2, and the yarns sit flatter — which is why twills drape better and pack more yarns per centimetre.

5. Pattern changes: add chest ease (widen the front panel) and/or lower the armhole / add a back yoke pleat so reaching forward does not pull the chest. The suspect cloth property is bending (and shear) stiffness: if the entered stiffness is far below the real fabric’s, the simulated cloth stretches and flows where the real garment would bind — the fit map flatters a garment that will actually pull.
6. Permeability is flow through the actual voids between yarns; closing them in the model zeroes the very quantity being computed and falsifies pressure-drop and filtration predictions. A sales render, by contrast, sells appearance — real pores at textile scale mostly read as noise or moiré, so the artist may tighten the weave visually for a clean, premium look. Same model family, different master: physics versus photography.
